



GTC4Lusso USA - Stop&Start system (for vehicles with HELE) - Latest Update: 2021-01-26

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F2.02 Stop&Start system (for vehicles with HELE)

Stop&Start system function



Set the vehicle in Key-OFF state before starting any maintenance work on vehicles with the Stop&Start system.

Before performing any maintenance procedures requiring the electronic system of the vehicle to be in Key-ON state on vehicles with the Stop&Start system, deactivate the Stop&Start system with the relative button, ensuring that the specific icon and the message "Stop&Start OFF" are shown on the TFT display, and that the LED on the on/off button itself is off.

Failure to do so may result in serious injury to persons working on the vehicle and in the vicinity of the vehicle in the event of the engine starting unexpectedly.

Vehicles with the optional HELE (High Emotion Low Emissions) packages are equipped with a Stop&Start system, which automatically switches off the engine when the vehicle is stationary and automatically restarts the engine when the driver intends to continue driving.

The purpose of Stop&Start is to increase the efficiency of the vehicle and reduce CO2 emissions, especially in urban use.

The automatic engine shut-off and restart functions are implemented with a series of sensors and control strategies devised to minimise the perceptible effects of the system on driveability and driving comfort and to keep all safety systems fully functional and maintain the desired climate comfort with the engine off.

When enabled, the Stop&Start system automatically stops and restarts the engine in relation to driving conditions, on board climate comfort and vehicle status.

The logic of the system does not require the driver to make any changes to the way he or she drives the vehicle.

Each time the ignition is turned to KEY-ON, the Stop&Start system will be in the same state it was in when the ignition switch was last turned to KEY-OFF.

Therefore, if the function was active at the last Key-off, it will also be active at the next Key-on.

If the engine continues running with the vehicle stationary and the brake pedal depressed even though Stop&Start is enabled, this means that the system has inhibited automatic engine shut-off. In this situation, non specific indication is displayed on the instrument panel: this is normal.

If automatic restart is permanently disabled, the vehicle behaves in exactly the same way as a car without Stop&Start: the engine must therefore be restarted manually, pressing the ENGINE START buttons and using the shift paddles behind the steering wheel as required.



The driver may activate or deactivate the Stop&Start system at any time by pressing button **(A)** on the roof panel.

If the Stop&Start system is active, the LED on the button **(A)** is off; the LED illuminated if the system is off. If the system is switched off by pressing the button **(A)** with ignition on and engine off (or if the engine has been stopped automatically by the Stop&Start system), the engine will restart automatically.



If the system is deactivated, press the button to activate: the message **"Stop&Start ON"** is displayed on the TFT display for 5 seconds, together with the specific symbol.

When the system is active, the engine stops automatically if the vehicle is stationary with the brake pedal depressed, and restarts when the brake pedal is released.

The following conditions must be met to enable automatic engine cut-out while the vehicle is running:

- Stop&Start system activated from button on roof panel (LED on);
- driver seat belt buckled.



The TFT display notifies the driver when the Stop&Start system has stopped the engine by displaying the message **"Stop&Start attivo"**, together with the specific symbol.

This message is displayed continuously for as long as the vehicle is in the stop state.

To cancel the message and reduce the symbol to icon form, press the MODE button ("ESCAPE" function).

For reasons of safety, to maintain comfort levels and to ensure that the vehicle and its individual systems function correctly, automatic engine cut-out is inhibited in the following cases:

- accelerator pedal depressed;
- when the system recognises a potentially hazardous situation (vehicle on a gradient, vehicle stationary with wheels steered - e.g. while vehicle is about to turn into a junction or a roundabout);
- when the system recognises that the vehicle is in a traffic queue or parking (by monitoring reverse gear state, time elapsed since last stop and vehicle speed);
- with maximum AC system output requested ("low", "max defrost" or "defrost rear screen");
- with very low or very high external temperatures;
- gearbox in "N";
- "LIFT" system moving;
- DCT gearbox Park Lock function released manually;
- in the event of a malfunction compromising the functionality of the system itself.

The system also constantly monitors a number of vehicle parameters (battery state of charge, engine coolant temperature, catalytic converter temperature, clutch oil temperature, 4WD system clutch oil temperature, brake servo vacuum level, climate comfort in passenger compartment) and will automatically inhibit engine shut-off if necessary to maintain safety, vehicle functionality and on-board climate comfort.

With the Stop&Start system active and the vehicle in stop state, performing any of the following actions will restart the engine:

- releasing the brake pedal;
- pulling the UP shift paddle;
- engaging reverse;
- pressing the accelerator pedal.

For reasons of safety, to maintain comfort while the vehicle is stationary and to ensure that the entire vehicle and its systems function correctly, the Stop&Start will automatically restart the engine in any of the following cases:

- ENGINE START button on steering wheel pressed;
- function deactivated by pressing button on roof panel;

- driver side seat belt unbuckled;
- driver side door opened;
- "LIFT" system activated;
- rear screen defrost requested or AC system set to "low" or "max defrost".

In certain cases, after the Stop&Start system has automatically stopped the engine, the engine may not restart automatically when requested by the driver: this means that the vehicle systems have permanently disabled automatic restart.



In this case, the engine must be restarted manually: the condition is indicated by the message **"Press ENGINE START button to restart"** on the TFT display, accompanied by an audible signal.

Automatic engine restart is permanently disabled in the following conditions:

- engine compartment lid opened;
- failed automatic engine restart;
- detection of a fault which may compromise the functions of the Stop&Start system.

In the event of a malfunction, the Stop&Start system is disabled.

The driver is warned of the fault by the specific symbol, displayed on the left hand TFT display for 20 seconds, accompanied by an audible warning signal and one of the following two messages, which correspond to two different alarm levels:

level 1 alarm: **"Stop&Start failure"**.

The message disappears at the end of the display cycle (20 seconds) or if the MODE button is pressed (with "ESCAPE" function), while the symbol remains, reduced to icon form.

level 2 alarm: **"Stop&Start failure. Go to dealer"**.

The message disappears at the end of the display cycle (20 seconds) or if the MODE button is pressed (with "ESCAPE" function), while the symbol remains, reduced to icon form.



Turn the vehicle off before starting any maintenance and/or repair work. If it is necessary to work on the vehicle with the instrument panel active, before starting any maintenance and/or repairs, check that the Stop&Start system is off. Check that the message **"Stop&Start OFF"** appears on the left hand TFT display for 5 seconds. The LED on the button must be lit.

Replacing Eberspeacher voltage stabiliser

			
Tightening torque		Nm	Class
Fastening for passenger side footrest mat mounting bracket	Screw	10 Nm	B

- Disconnect the battery (🔧 F2.01).
- Remove covers and mats (🔧 E4.05).
- Remove the passenger side footrest mat.



- Undo the screws indicated.
- Remove the bracket (A).



- Disconnect the connector (1).
- Undo the nuts indicated.
- Remove the voltage stabiliser (B) and replace.
- Fit the new voltage stabiliser (B).
- Hand tighten the nuts indicated.
- Connect the connector (1).



- Fit the bracket (A).
- Tighten the screws indicated.

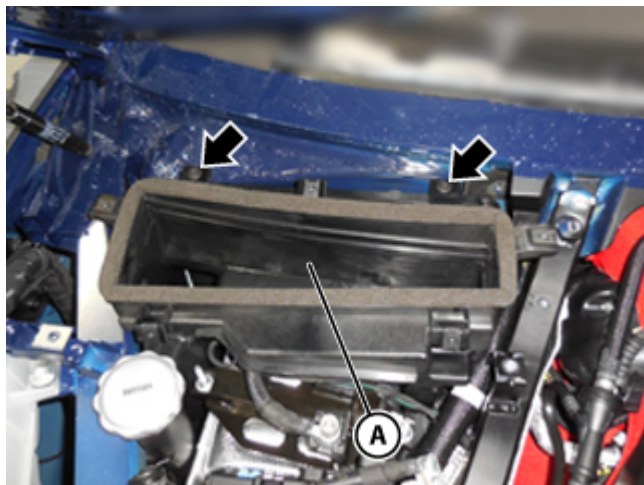
		
Tightening torque	Nm	Class
Screw	10 Nm	B

- Refit covers and mats (🔧 E4.05).
- Refit the passenger side footrest mat.
- Connect the battery (🔧 F2.01).

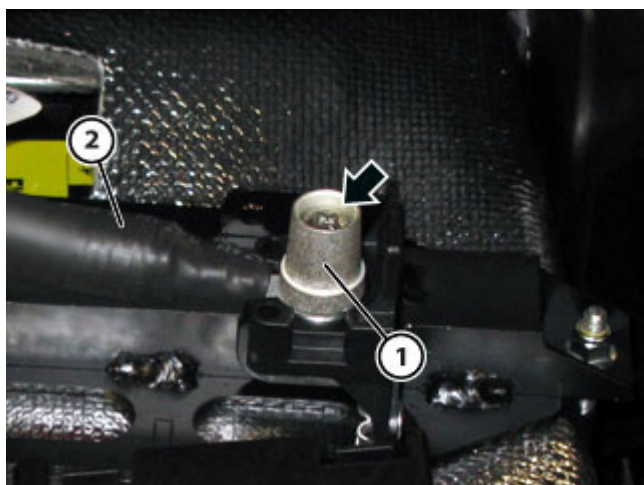
Replacing the battery sensor

			
Tightening torque		Nm	Class
Fastening battery sensor	Nut	6 Nm	B
Fastening the passenger compartment air intake	Screw	4.5 Nm	B

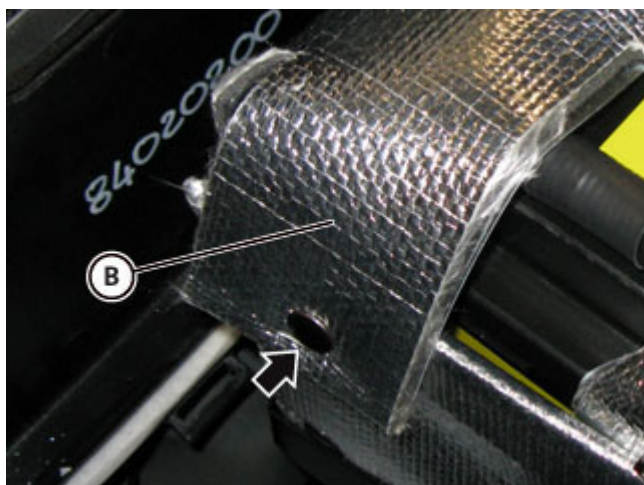
- Disconnect the battery (🔧 F2.01).
- Remove the engine compartment cosmetic shields (🔧 E3.14).
- Remove the right hand underwindscreen shield.



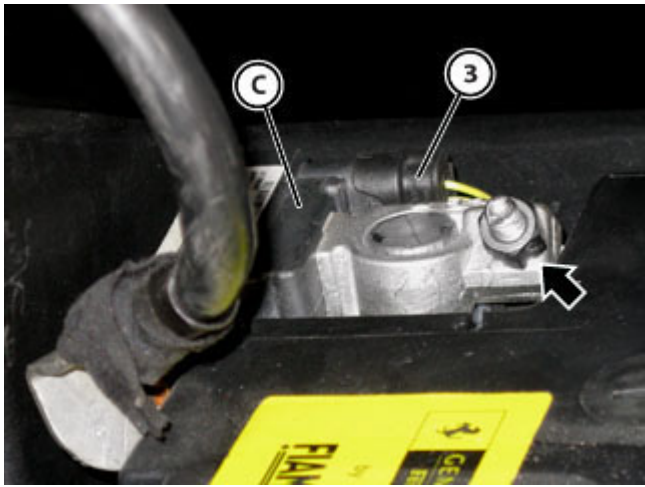
- Undo the screws indicated.
- Remove the passenger compartment air duct (A).



- Remove the false negative pole (1) and disconnect the cable (2), undoing the indicated screw.

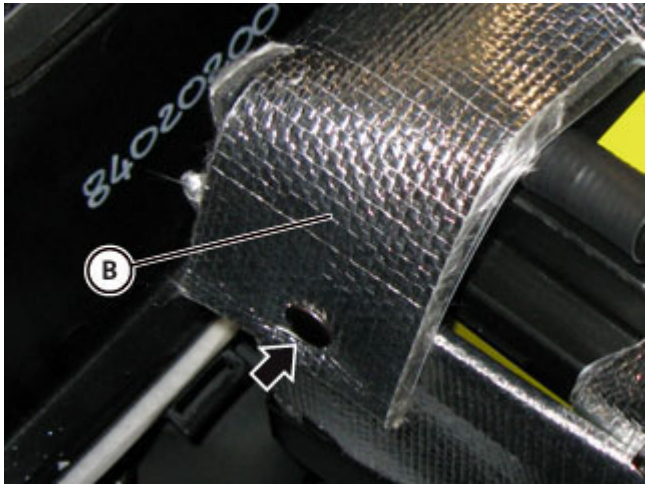


- Lift the protective cover (B), detaching the indicated clip.

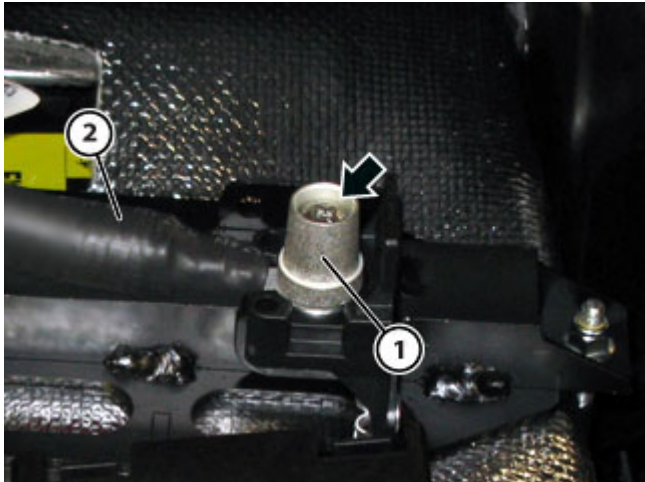


- Disconnect the connector **(3)**.
- Loosen the indicated nut.
- Remove the sensor **(C)** from the negative battery pole and replace.
- Clean and degrease the inner surface of the new battery sensor terminal and the surface of the negative battery pole.
- Fit the sensor **(C)** on the negative battery pole.
- Tighten the indicated nut.

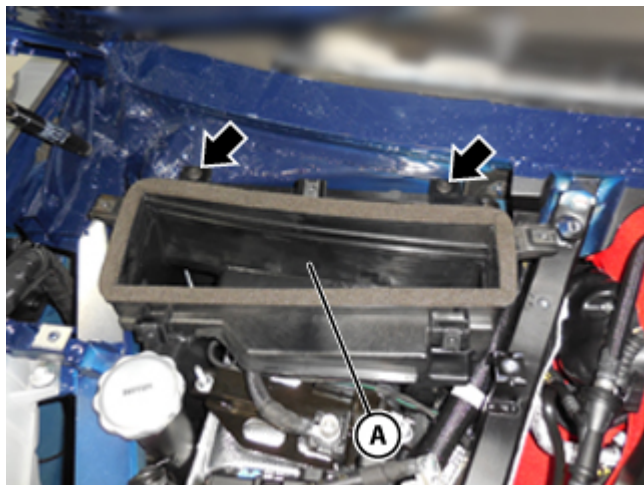
		
Tightening torque	Nm	Class
Nut	6 Nm	B



- Connect the connector **(3)**.
- Lift the protective cover **(B)**, fastening the indicated clip.



- Connect the battery sensor cable **(2)** and fit the false negative pole **(1)**, tightening the indicated screw.



- Fit the passenger compartment air duct **(A)**.
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	4.5 Nm	B

- Refit the engine compartment cosmetic shields ( E3.14).
- Connect the battery ( F2.01).
- Perform battery sensor calibration ( F2.02).

Battery sensor calibration

When the battery sensor is powered for the first time, after battery replacement or after an interruption in the power supply, it runs a calibration cycle, during which it recognises the type of battery, its characteristics and its state of charge.

For the Stop&Start system to function correctly, the battery sensor calibration cycle must be completed correctly.

- Start the engine.
 - Switch off the engine
-  DO NOT connect the vehicle to the battery conditioner.
DO NOT disconnect the battery.

- Leave the vehicle stationary with the engine off for at least **6 hours**.
- Drive the vehicle for at least **20 minutes**.